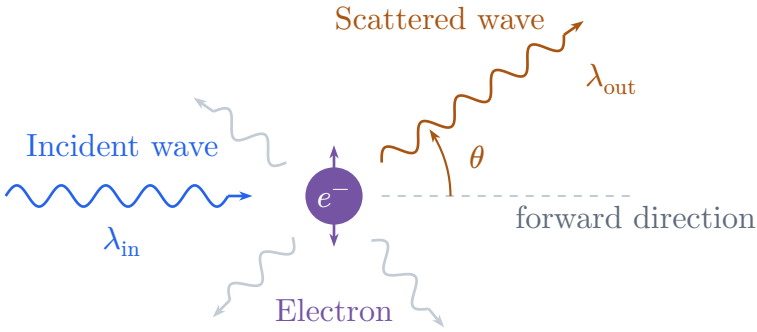
Thomson scattering

Elastic scattering by a free electron



$$\lambda_{\text{out}} = \lambda_{\text{in}} \quad h\nu_{\text{out}} = h\nu_{\text{in}}$$

The electric field drives the electron; the accelerated charge reradiates.

Classical limit: photon energy $h\nu \ll m_e c^2$; recoil is neglected.